* Standardization - Intution:

Also Known as 'z-score Normalization

Now lets take input column Age and Salary

Age (x_i)		Salary
32	x_1	84K
48	x_2	72K
15	x_3	19K
28	:	50K
34	:	61K
:		:

Now we have to Standardize Age So we make new column Age'

Now to transform Value we use formula

$$x_i' = \frac{x_i - \bar{x}}{\sigma}$$

where x_i = input column
$\bar{x}$ = mean
σ = standard deviation

After transforming the column, the μ and σ of that new column will be

$$\mu = 0, \sigma = 1$$

mean

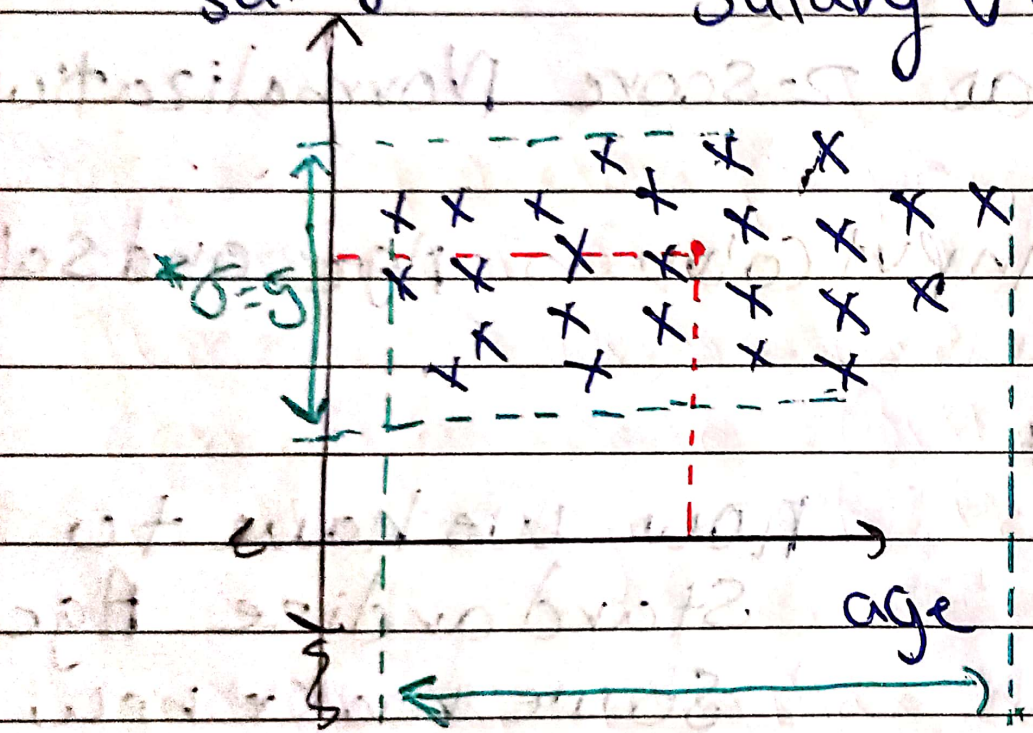
* sample data not exact calculation

— range or data spread

— mean

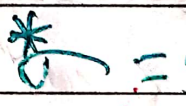
Geometry

So basically this is a graph of Salary v age before transformation

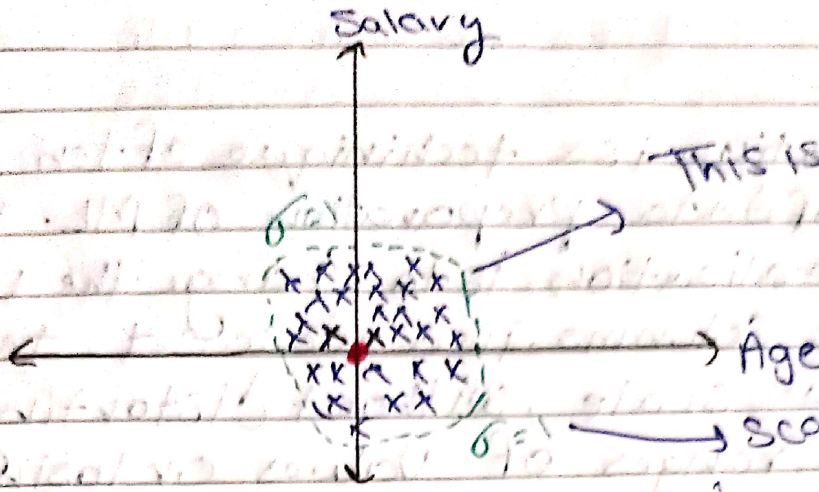


After Standardization

After transformation



Scanned with OKEN Scanner



Here the mean is 0 and Standard deviation is 1.

After transformation / Standardization which scale the value for ml algorithm by bringing in range

When to use Standardization :

i) Kmeans

ii) K-nearest-neighbours

iii) Principal Component analysis (PCA)

iv) Artificial neural network

v) Gradient Desent